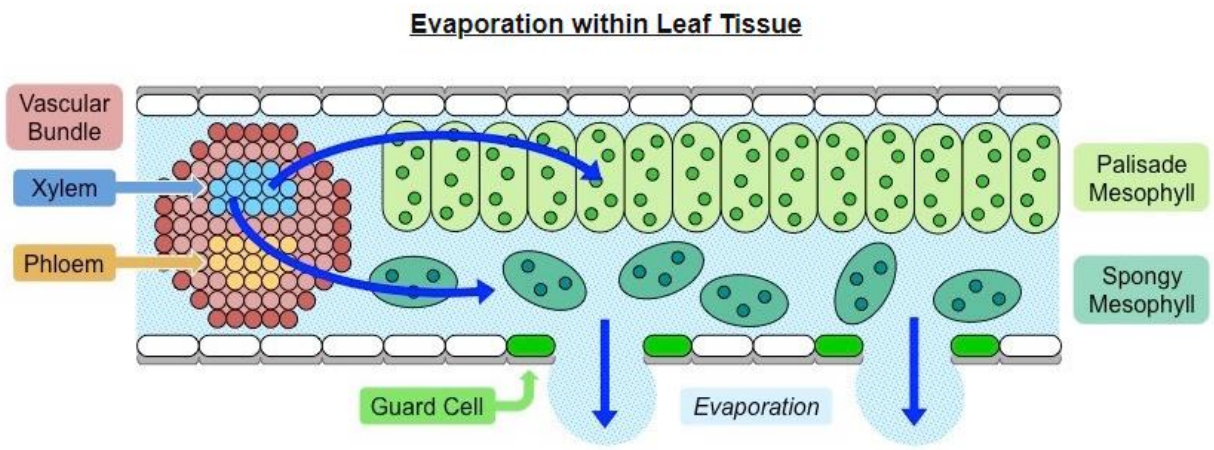
Ch 10 Transpiration, transport and support in plants

10.1 Transpiration *(Book 1B, p. 10-2)*

- (1) _____ (蒸騰) is the **loss** of water in the form of **water vapour** from plant surfaces due to **evaporation**.

B Process of transpiration *(Book 1B, p. 10-4)*



❶ Water diffuses out of the mesophyll cells to form a (5) _____ _____ on the cell surface.

❷ Water of the water film (6) _____ to form **water vapour**. The water vapour moves into the **air space** among the mesophyll cells.

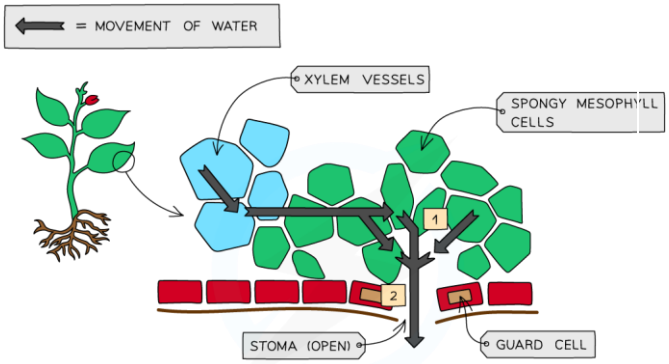
❸ Water vapour in the air space (7) _____ through the stomata **to the atmosphere**.

Creation of transpiration pull

ⓐ During transpiration, mesophyll cells near the air space lose water to the air space. Their **water potential** (8) _____ (lowers / rises).

ⓑ Water moves from neighbouring cells into these cells by (9) _____. This is repeated across the layer of mesophyll cells.

ⓒ Eventually, water moves out of the (10) _____ _____ by osmosis to replace the water loss in mesophyll cells. **Transpiration pull** (蒸騰拉力) is created to pull water up the xylem vessels from roots.



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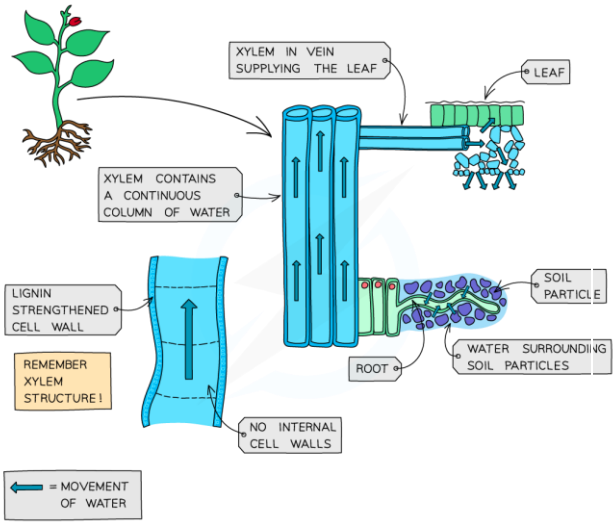
C Significance of transpiration *(Book 1B, p. 10-5)*

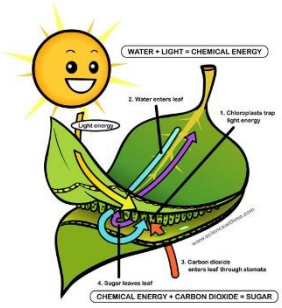
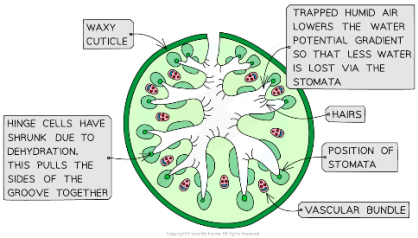
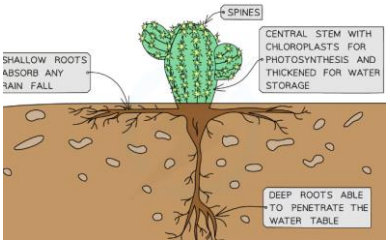
1 Transpiration **creates transpiration pull**, which enables (11) _____ **and** (12) _____ to be **transported** from the roots to other parts of the plant.

2 Transpiration pull **facilitates the** (13) _____ **of water** by roots.

3 During transpiration, the evaporation of water from the mesophyll cells removes (14) _____ from the leaves. This produces a **cooling effect** to prevent plants from being (15) _____ under hot conditions.



D Adaptations of leaves to prevent excessive water loss *(Book 1B, p. 10-6)*

Feature of leaves of terrestrial plants	Adaptation
Epidermis **covered with a waxy** (16) _____	Cuticle is almost impermeable to water. This reduces water loss through leaf surfaces.
Few or no stomata on the (17) _____ (upper / lower) **epidermis** of most dicotyledonous leaves	
(18) _____ _____ are present to **control the opening and closing of stomata**	At night, the need for gas exchange in plants decreases. Most stomata (19) _____ (close / open) to reduce water loss from the leaves.
	

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E Distribution of stomata on leaves of different types of plants

(Book 1B, p. 10-8)

Type of plant	Distribution of stomata on leaves and their adaptation
Terrestrial dicot plants	**Lower epidermis has** (20) _____ (fewer / more) **stomata** than the upper epidermis. This can help reduce water loss.
Terrestrial monocot plants	The stomata are almost (21) _____ **distributed on both sides** of the leaves. As the leaves are **oriented vertically**, both sides of the leaves receive similar sunlight. They lose water at a (22) _____ rate.
Submerged plants	Stomata are **absent** in the epidermis.
Floating plants	Stomata are **present in the** (23) _____ (upper / lower) **epidermis only**.

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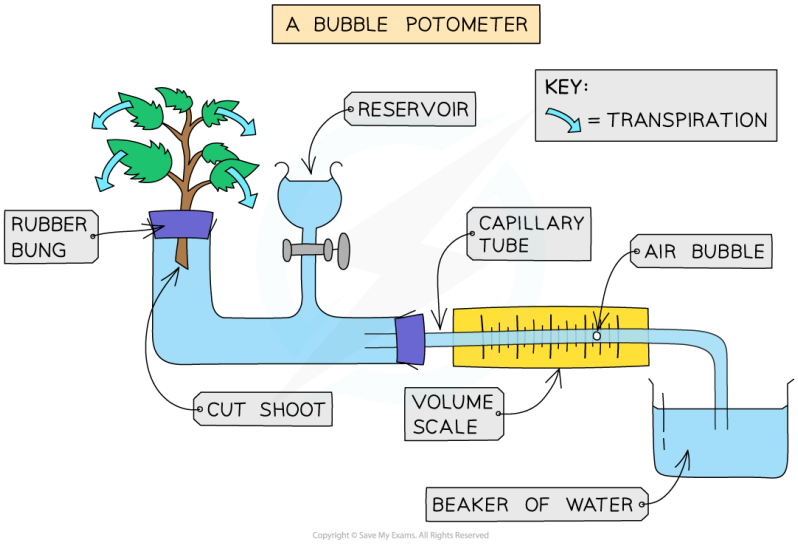
- A (24) _____ (蒸騰計) can be used to estimate the **rate of transpiration** of a plant. There are different types of potometers. Below are two examples:
  - (25) _____ **potometer** (氣泡蒸騰計):

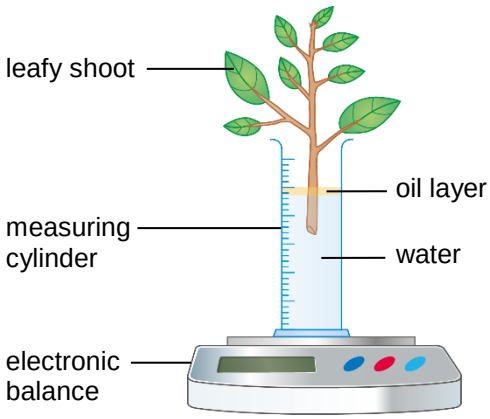
    It can measure the **rate of water uptake** by a plant. It gives an indirect measurement of the rate of transpiration by **assuming that**

    (25a) _____.



  - (26) _____ **potometer** (重量蒸騰計):

    The measuring cylinder is used to **measure the amount of water** (27) _____ (absorbed / lost) by a plant. The electronic balance is used to **measure the amount of water** (28) _____ (absorbed / lost) by the plant due to transpiration.



The set-up of a weight potometer

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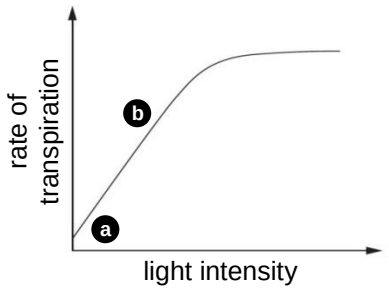
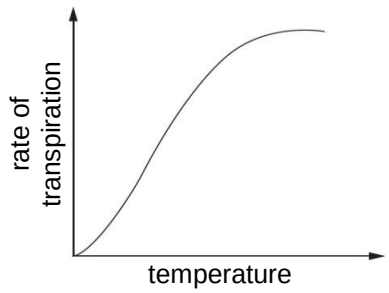
G Environmental factors affecting the rate of transpiration *(Book 1B, p. 10-13)*

- **Light intensity**, (29) _____, **air movement** and **relative humidity** are some of the environmental factors that affect the rate of transpiration of a plant.

Light intensity	
	**a** The rate of transpiration is low in darkness. **Reason**: The **stomata are** (30) _____. Only a small amount of water vapour diffuses out of the leaves. **b** The rate (31) _____ when the **light intensity increases**. **Reason**: As light intensity increases, the **stomata open** (32) _____. The cross-sectional area for the diffusion of water vapour (33) _____. Water vapour diffuses out of the leaves more rapidly.

Temperature	
	■ The **rate increases** when the **temperature** (34) _____. **Reason**: As temperature increases, the **rate of** (35) _____ **of water** from the surfaces of mesophyll cells **increases**. This increases the concentration gradient between the air space and the surrounding air. As a result, water vapour diffuses out of the leaves more rapidly.

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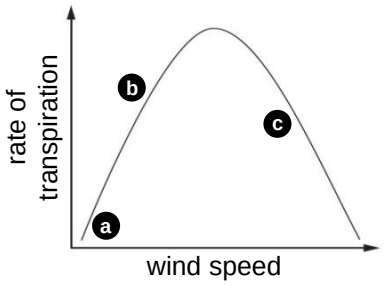
Air movement



a The rate of transpiration is low in still air.

Reason: **Water vapour** (36) _____ around the stomata. This **decreases the concentration gradient of water vapour** between the air space and the surrounding air. The rate of diffusion of water out of the leaves decreases.

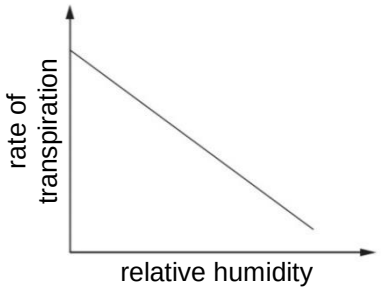
b The **rate increases** when the **wind speed increases**.

Reason: Wind blows away water vapour around the stomata. This helps **maintain a steep** (37) _____ _____ **of water vapour** between the air space and the surrounding air. Water vapour diffuses out of the leaves more rapidly.

c The rate (38) _____ when the wind is too strong.

Reason: Under strong wind, water may be lost from the plant too rapidly, causing **most stomata to** (39) _____. Less water vapour diffuses out of the leaves.

Relative humidity



- The rate (40) _____ when the **relative humidity** of the surrounding air **increases**.

  Reason: A **higher relative humidity** of the surrounding air (41) _____ **the concentration gradient of water vapour** between the air space and the surrounding air. Less water vapour diffuses out of the leaves.

- Besides environmental factors, the structure of the plant (e.g. surface area of leaves, (42) _____ of the cuticle and abundance of (43) _____) also affects the rate of transpiration.

Practical 10.7 Design an investigation of the effects of environmental factors on the rate of transpiration

(Book 1B, p. 10-13; Practical Workbook for SBA 1B, p. 10-21)

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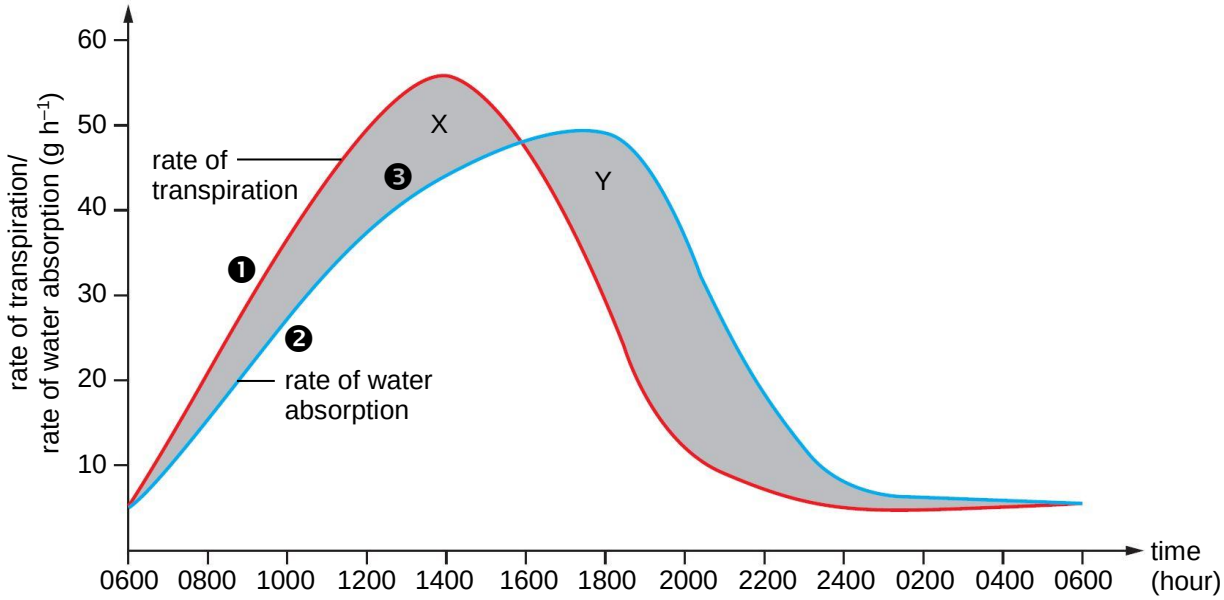
Relationship between rates of transpiration and water absorption of plants



The changes in the rates of transpiration and water absorption of a plant in 24 hours

❶ The rate of transpiration increases from 0600 to 1400 hours.

- Light intensity increases in this period. This causes the stomata to open (44) _____.
- Temperature also rises in this period. This increases the rate of (45) _____ of water.
- As a result, water vapour diffuses out of the leaves more rapidly.

❷ The curves of water absorption and transpiration are similar in shape.

- They are interdependent. An increased rate of transpiration results in a larger (46) _____ _____, which in turn causes more water to be absorbed from the soil.

❸ Area **X** represents the **net amount of water** (47) _____ (absorbed / lost) by the plant from 0600 to 1600 hours. Area **Y** represents the **net amount of water absorbed** by the plant from 1600 to 0600 hours.

- Area Y is (48) _____ (smaller / larger) than area X. This suggests that the plant has a **net** (49) _____ (uptake / loss) **of water** during the 24-hour period.
- The water retained is likely being used for photosynthesis, formation of new cells and other metabolic activities.

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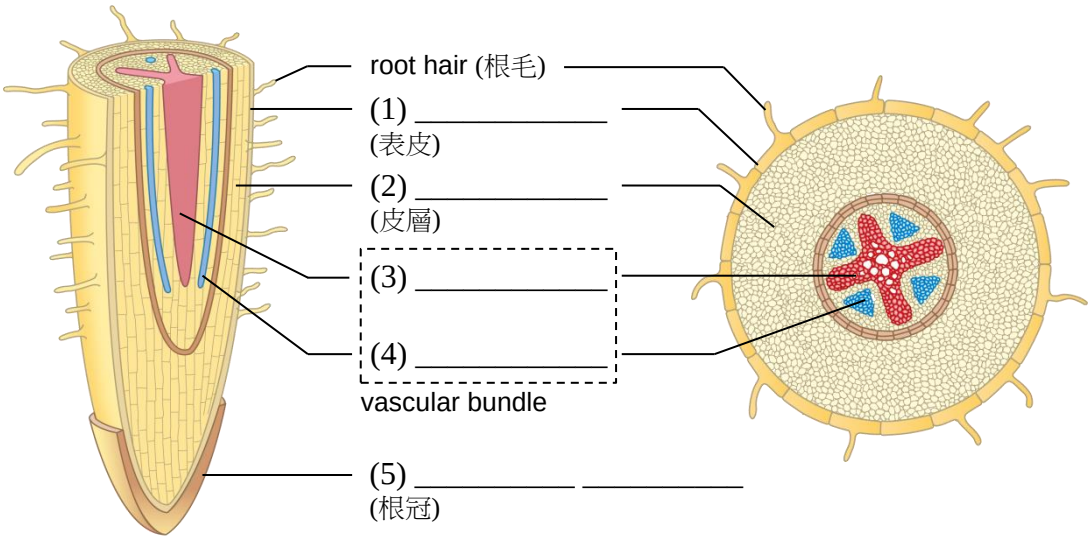
10.2 Absorption of water and minerals by roots *(Book 1B, p. 10-18)*

A Structure of the root *(Book 1B, p. 10-18)*



Longitudinal section of a dicotyledonous root

Transverse section of a dicotyledonous root

Structure	Feature and function
Epidermis	■ **Not covered by** (6) _____ ■ Made up of a layer of **closely-packed thin-walled cells** ■ **Protects** the inner tissues from injury and (7) _____ ■ Some cells have (8) _____ _____ which provide a **large** (9) _____ _____ for the absorption of water and minerals
Cortex	■ Made up of layers of **loosely-packed thin-walled cells** ■ **Stores food** in the form of (10) _____ ■ **Allows** the **passage of water and minerals** across the root
Vascular bundle	■ **Xylem transports** (11) _____ and (12) _____ ■ **Phloem transports** (13) _____ _____
Root cap	■ Covers and **protects** the **root tip** ■ Constantly worn away and replaced by new cells produced in the root tip

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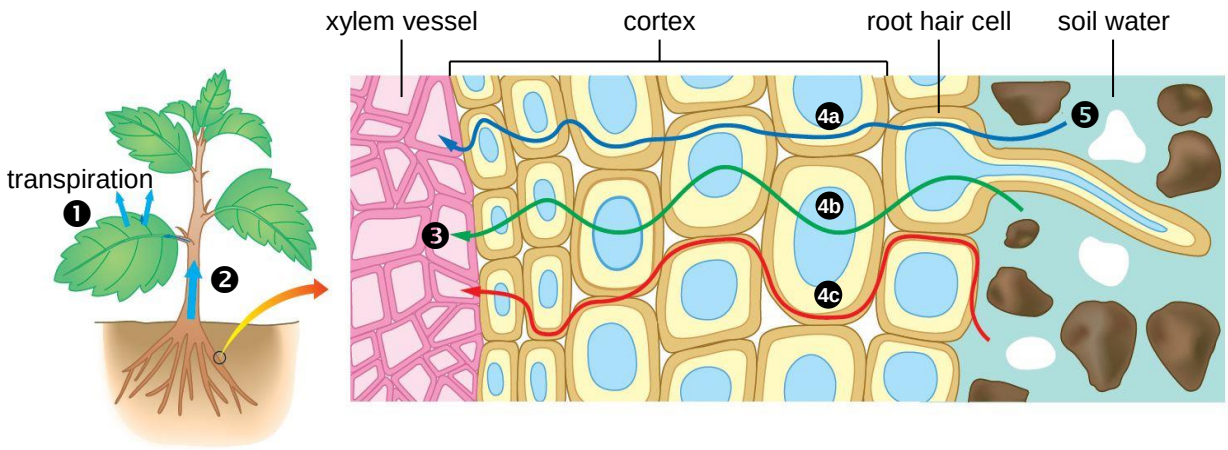
B Process of absorption of water by roots *(Book 1B, p. 10-20)*



The process of absorption of water by the root

1. Water is **lost** from the leaves continuously by (14) _____. This creates transpiration pull.
2. Water is **drawn up the** (15) _____ _____ from the roots to the leaves by the **transpiration pull**.
3. Water in the cortex cells near the xylem vessels of the roots enters the xylem vessels. This decreases the water potential of them. A (16) _____ _____ _____ is set up across the whole cortex.
4. In the cortex, water travels from cell to cell through three routes:
   - a. Water moves along a water potential gradient through the (17) _____ of cells by **osmosis**.
   - b. Water moves along the same water potential gradient through the (18) _____ of cells by **osmosis**.
   - c. Water travels from one cell to another freely through the (19) _____ _____.
5. As water is drawn away from the root hair cells to the neighbouring cortical cells, the water potential of the root hair cells becomes lower. As the water potential of the soil water is (20) _____ (lower / higher) than that of the root hair cells, water in the soil enters the root hair cells by (21) _____.

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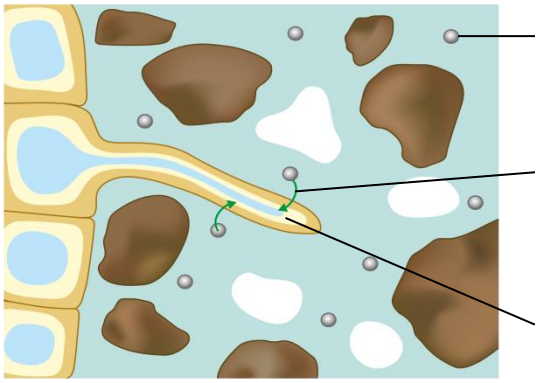
C Process of absorption of minerals by roots *(Book 1B, p. 10-21)*

- The concentrations of minerals in the soil are usually lower than that in the root hair cells. Minerals are mainly absorbed **against the concentration gradient** by (22) _____ _____.
- Active absorption of minerals by root hair cells (23) _____ (lowers / raises) the water potential of the cells. This helps the absorption of water into the cells by (24) _____.



➤ Absorption of minerals into the root

- On rare occasions, when the concentrations of minerals in the soil are higher than that in the root hair cells, minerals enter the root hair cells down the concentration gradient by (25) _____.

D Adaptive features of roots for absorption of water and minerals *(Book 1B, p. 10-22)*

Feature	Adaptation
Highly branched and have **numerous** (26) _____ _____	Provide a large (27) _____ _____ for the absorption of water and minerals
Long and **fine root hairs**	Allows root hairs to grow between soil particles to absorb water and minerals
Epidermis **not covered by** (28) _____ and made up of **one layer of thin-walled cells** only	Allows water and minerals to pass through the epidermis into the root easily
Many (29) _____ in root hair cells	Ensures enough energy is supplied to absorb minerals from the soil by active transport

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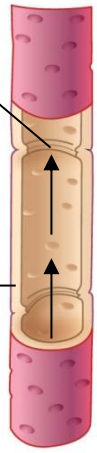
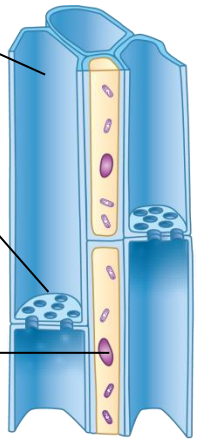
10.3 Transport in plants *(Book 1B, p. 10-24)*

A Types of vascular tissues *(Book 1B, p. 10-24)*

(1) _____	
Function	■ **Transports water** and **minerals** ■ **Provides support** for plants
Adaptive feature	■ Mainly consists of (2) _____ _____ (木質導管) which are **hollow tubes** made up of (3) _____ (living / dead) **cells**. **1 Hollow tube (with no cell contents and no end walls between cells)** ■ allows the passage of water with little resistance **2 Cell walls are thick and** (4) _____ (木質化的) ■ prevents vessels from collapsing under the great tension of transpiration pull 

(5) _____	
Function	■ **Transports organic nutrients** (mainly sugars)
Adaptive feature	■ Mainly consists of (6) _____ _____ (篩管) and **companion cells** (伴細胞). Sieve tubes are elongated (7) _____ (living / dead) **cells** joined end to end. **1 Sieve tube contains little cytoplasm and no** (8) _____ ■ allows organic nutrients to move along with little resistance **2 Sieve plate** (篩板) **has** (9) _____ ■ allows organic nutrients to pass through **3 Companion cell has dense cytoplasm and many organelles** (including nucleus) ■ supports metabolism of the sieve tube 

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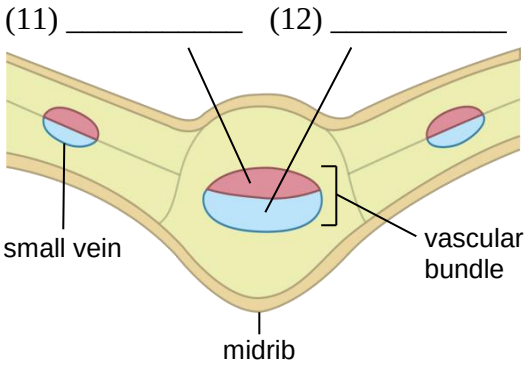
B Distribution of vascular tissues *(Book 1B, p. 10-26)*

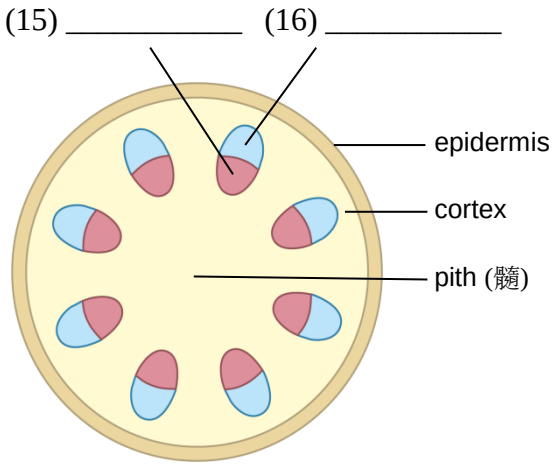
- In flowering plants, vascular tissues are arranged in long strands called (10) _____ _____.
- Distributions of vascular bundles in the root, stem and leaf in a young dicotyledonous plant:

Leaf:



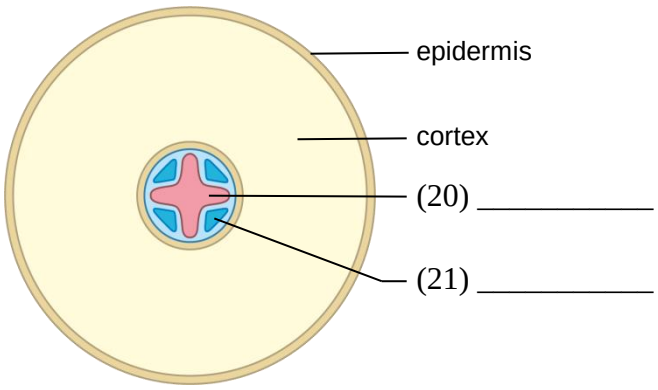
- Vascular bundles are found in the large central (13) _____ and the network of small **veins**.
- **Xylem lies on** (14) _____ of phloem.

Stem:



- Vascular bundles are arranged in a **ring** at the (17) _____.
- (18) _____ (Xylem / Phloem) is located in the **inner** region and (19) _____ (xylem / phloem) in the **outer** region.

Root:



- Vascular bundles are located at the (22) _____.
- **Phloem** is found **between** the 'arms' of **xylem**.

Practical 10.8 Examination of the vascular tissues of a young dicotyledonous plant
(Book 1B, p. 10-27; Practical Workbook for SBA 1B, p. 10-27)

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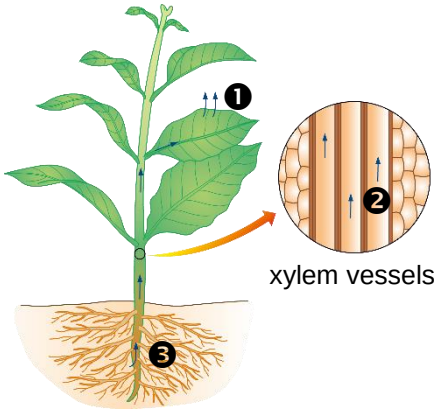
C Process of transport of water and minerals *(Book 1B, p. 10-28)*



Water and minerals are transported up the xylem vessels in plants

❶ Water is lost from leaves through (23) _____. Transpiration pull is created.

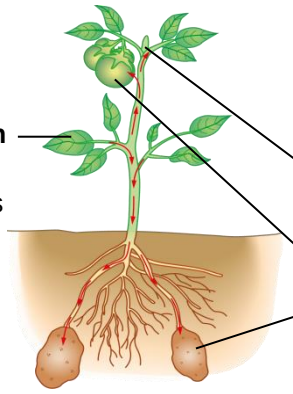
❷ Water (with dissolved minerals) is drawn up the (24) _____ _____ by transpiration pull as a **continuous stream**.

❸ Water in the soil is absorbed into the root by (25) _____. Dissolved minerals are absorbed by (26) _____ _____.

Practical 10.9 Identifying the vascular tissue in the stem responsible for water transport
(Book 1B, p. 10-29; Practical Workbook for SBA 1B, p. 10-33)

D Translocation of organic nutrients in plants *(Book 1B, p. 10-30)*

- Organic nutrients (carbohydrates) are synthesized in the leaves by photosynthesis and then transported by (27) _____ to other parts of the plant where they are used or stored.
- The transport of organic nutrients in plants is called (28) _____ (輸導). It is an (29) _____ **process** (i.e. **requires energy**).



Translocation of organic nutrients in plants

Skill builder

Drawing low-power biological diagrams *(Book 1B, p. 10-33)*

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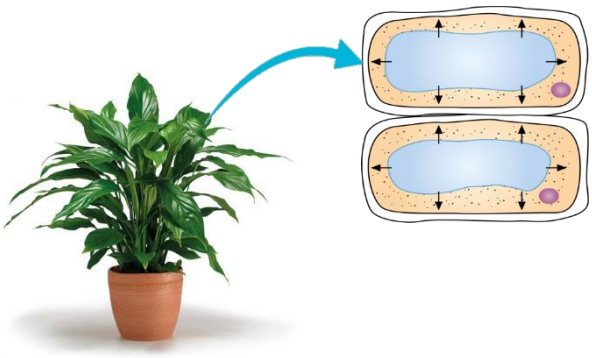
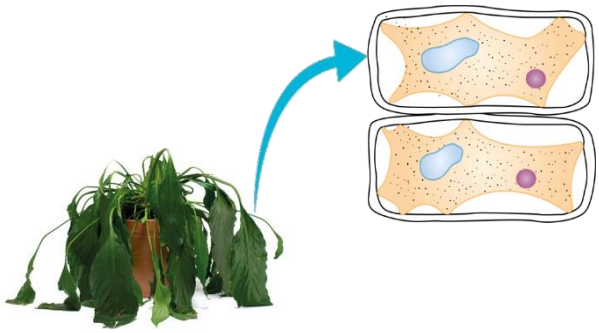
10.4 Support in plants *(Book 1B, p. 10-35)*

- Importance of support (支持) to terrestrial plants:
  - To hold the **leaves** in the best position to **receive the maximum amount of** (1) _____ for photosynthesis
  - To lift up the **flowers** to **facilitate** (2) _____
  - To lift up the **fruits** and **seeds** to **facilitate** their (3) _____
- **Support** in plants is mainly provided by the (4) _____ **of thin-walled cells** and the (5) _____ **of thick-walled cells**.

A Turgidity of thin-walled cells *(Book 1B, p. 10-35)*

- **Turgidity** of **thin-walled cells** provides support to **leaves**, **herbaceous stems** and other **non-woody** plant parts of a plant:

Adequate water supply	Inadequate water supply
	
▪ Thin-walled cells in the stems and leaves **gain water** by **osmosis**. They become (6) _____ and press against each other. This gives support to the plants. ▪ The plant **stands upright**.	▪ Thin-walled cells in the stems and leaves **lose water**. They become (7) _____ and can no longer support the plant. ▪ The plant **wilts**.

B Rigidity of thick-walled cells *(Book 1B, p. 10-36)*

- (8) _____ _____ and the tissue outside each vascular bundle in young dicot stems consist of **thick-walled cells**. Their cell walls contain (9) _____, which makes the cells **hard** and **rigid**. The rigidity of these cells supports the plant.
- When woody plant grows, **more xylem** is formed in **woody stems** and more lignin is deposited in the cell walls of the older xylem. The older xylem finally becomes (10) _____, which provides great support to the plants.

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10.5 Comparison of transport in humans and flowering plants

(Book 1B, p. 10-38)

		Humans	Flowering plants
Structure of transport system	Similarities	**Two systems of vessels** are involved: In humans: (1) _____ vessels and lymph vessels In plants: xylem and (2) _____	
	Differences	Transport of water:	
		Nature of the cell:	
		Texture of the vessels	Xylem vessels are relatively (3) _____.
		Prevent back flow:	**(3a)** _____ are present in sieve tubes.
Mechanism of transport	Similarities	A **driving force** is provided to move substances along the vessels.	
	Differences	Force come from:	**No special organ** is developed to generate a driving force.
		The elastic walls of arteries (5) _____ to move blood forward.	The walls of xylem vessels and sieve tubes are inelastic and cannot recoil.
		Valves are present in veins and lymph vessels to **prevent** (6) _____ of blood and lymph.	No special structures in xylem and phloem to prevent backflow of fluid.
		Contraction of (7) _____ _____ squeezes blood and lymph forward.	No contraction of tissues in plants.

Skill builder

Answering essay questions *(Book 1B, p. 10-39)*

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Answers

Ch 10 Transpiration, transport and support in plants

10.1

1	Transpiration	**2**	stomata	**3**	cuticle	**4**	lenticels
5	water film	**6**	evaporates	**7**	diffuses	**8**	lowers
9	osmosis	**10**	xylem vessels	**11**	water / minerals	**12**	minerals / water
13	absorption	**14**	heat	**15**	overheated	**16**	cuticle
17	upper	**18**	Guard cells	**19**	close	**20**	more
21	equally	**22**	similar	**23**	upper	**24**	potometer
25	bubble	**26**	weight	**27**	absorbed	**28**	lost
29	temperature	**30**	closed	**31**	increases	**32**	wider
33	increases	**34**	increases	**35**	evaporation	**36**	accumulates
37	concentration gradient			**38**	decreases	**39**	close
40	decreases	**41**	decreases	**42**	thickness	**43**	stomata
44	wider	**45**	evaporation	**46**	transpiration pull	**47**	lost
48	larger	**49**	uptake				

10.2

1	epidermis	**2**	cortex	**3**	xylem	**4**	phloem
5	root cap	**6**	cuticle	**7**	infections	**8**	root hairs
9	surface area	**10**	starch	**11**	water / minerals	**12**	minerals / water
13	organic nutrients	**14**	transpiration	**15**	xylem vessels		
16	water potential gradient			**17**	cytoplasm	**18**	vacuoles
19	cell wall	**20**	higher	**21**	osmosis	**22**	active transport
23	lowers	**24**	osmosis	**25**	diffusion	**26**	root hairs
27	surface area	**28**	cuticle	**29**	mitochondria		

10.3

1	Xylem	**2**	xylem vessels	**3**	dead	**4**	lignified
5	Phloem	**6**	sieve tubes	**7**	living	**8**	nucleus
9	pores	**10**	vascular bundles	**11**	xylem	**12**	phloem
13	midrib	**14**	top	**15**	xylem	**16**	phloem

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17	periphery	**18**	Xylem	**19**	phloem	**20**	xylem
21	phloem	**22**	centre	**23**	transpiration	**24**	xylem vessels
25	osmosis	**26**	active transport	**27**	phloem	**28**	translocation
29	active						

10.4

1	sunlight	**2**	pollination	**3**	dispersal	**4**	turgidity
5	rigidity	**6**	turgid	**7**	flaccid	**8**	Xylem vessels
9	lignin	**10**	wood				

10.5

1	blood	**2**	phloem	**3**	rigid	**4**	Valves
5	recoil	**6**	backflow	**7**	skeletal muscles		

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